

CompTIA Server+

Server Licensing Concepts

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Introduction

Server+

Licensing

Server License

Socket License

Core License

Open Source License

Compatibility

Welcome to the **Server Licensing Concepts** Practice Lab. In this module, you will be provided with information on server licensing.

Licensing is important to define the terms and conditions for the use of the product, end-user license agreements, and the service level agreements of support. In this module, you will look at the different licensing models and concepts that can be applied to our servers and discuss some of the pros and cons of each model. There is no one-size-fits-all when it comes to licensing because organizations will have unique needs from each other, so licensing options will be available.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Licensing Models
- Exercise 2 - Version Compatibility

After completing this module, you should have further knowledge of:

- Physical Vs. Virtual
- Site License
- Per-Server License
- Per-Socket License
- Per-Core License
- Per-Instance License
- Volume Licensing
- Concurrent License
- Node-Locked License
- Open Source License
- Subscription License
- License Vs. Maintenance and Support
- License Count Validation - True Up
- Backward Compatibility
- Forward Compatibility

Exam Objectives

The following exam objectives are covered in this module:

2.8 Explain licensing concepts

- Models
- Open source
- Subscription
- License vs. maintenance and support
- Volume licensing
- License count validation
- Version compatibility

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

This module contains supporting materials for Server+ (SK0-005).

Click **Next** to proceed to the first exercise.

Exercise 1 - Licensing Models

There are different options of licensing models available for users and businesses to select when looking for a service. Some models will cost more than other models but may have additional features, capabilities, or support that the non-paid models have. Each organization must assess its infrastructure, determine its software requirements, and purchase the appropriate licenses. If you don't, you can use software illegally and risk fines and penalties for copyright infringement. It is critical to keep track of these licenses and to have a zero-tolerance policy for unlicensed software.

In this exercise, you will learn about these models and how they apply to servers.

Learning Outcomes

After completing this exercise, you should have further knowledge of:

- Physical Vs. Virtual
- Site License
- Per-Server License
- Per-Socket License
- Per-Core License
- Per-Instance License
- Volume Licensing
- Concurrent License
- Node-Locked License
- Open Source License

- Subscription License
- License Vs. Maintenance and Support
- License Count Validation - True Up

Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

Physical Vs. Virtual

Microsoft offers us options when it comes to licensing for physical servers, and these options vary based on the requirements of the organization. Windows Server 2022, 2019, and 2016 will have three different levels of service:

Essentials Edition

The Essentials Edition will be for small businesses. It can be used for twenty-five users and fifty devices and does not require client access licenses for users to access the server. This license is considered a **specialty server license**, so it is licensed Per-Server.

Standard Edition

The Standard Edition will be most suitable for medium to larger size businesses that have limited virtualization. This is considered a **core-based licensing model** and does not consider the number of users and devices accessing it. Instead, it looks at the number of cores the servers are using. There is a minimum of sixteen cores and additional costs for more cores. To account for the number of users and devices accessing the server, client access licenses will have to be purchased, which was not required in the Essentials. This model has a limit to virtualization, only allowing two virtual machines.

Datacenter Edition

The Datacenter Edition will be for large businesses that intend to utilize virtualization. This is a core-based licensing model like the standard edition and also requires client access licenses for users accessing the server. The main difference between the data center and the standard model is the data center model allows for unlimited virtual machines. This can help to reduce overall costs in the organization and maximize the use of the hardware available. This model will also support unlimited storage replication, software-defined networking, and software-defined

storage, which are not available in the standard edition. The 2022 data center model is the most expensive of the three.

Virtualized Servers can provide the same services as physical servers; in most cases, they will utilize a server and network hosted at a data center. After that, users can access the resources and services over the internet. Using a virtualized server hosted in the cloud enables businesses to save money on the cost of their original investment in hardware and the infrastructure needed to support their operations. Since no new wire will be required, both the wiring and the installation will be less expensive. The cloud service provider will be responsible for the upkeep and maintenance of the servers. Both their responsibilities are the cost of the cloud provider's electricity to power the server and the cost of the HVAC system to cool the space in which the server is housed.

Servers in a virtual environment offer the flexibility to expand or contract in response to the organization's needs. If they have a spike in business, they can rapidly deploy a new server and get it operational within a short amount of time. The server can then be turned off when it is no longer required, at which point it will no longer cost the business anything. The users of the service will be held accountable for their own data and will be in charge of monitoring how their equipment is utilized.

The cloud operates on a pay-as-you-go model, and if the system is not controlled effectively, the prices might quickly spiral out of control. It is also easy for companies to quickly rack up expenditures that are significantly higher than anticipated if employees forget to turn off machines that do not require their attention. Because of this, it is essential to have efficient management in the cloud.

Site License

Site Licenses, which are becoming increasingly rare, are a type of software and computing licensing that enables a single copy of the program to be at a single physical site. This also refers to physical hardware for our servers as well. These licenses date back to a period when the use of computers in business was far less common, and computers lacked the level of sophistication and portability that they have today. They were designed to simplify things by making it simpler to keep track of licenses and bring down associated costs.

Two often used regulations are put into practice: the maximum number of concurrent connections and the maximum number of devices that can run the software. Audits are conducted to make sure a license is being used as intended.

Per-Server License

Per-Server License refers to software licensing that permits installation on a server. It can include the number of concurrent users and the number of CPU cores. Microsoft offers multiple licensing tiers for its server software, each with various conditions and capabilities for its use.

Microsoft Server Essentials

Microsoft Server Essentials was created with the needs of smaller businesses in mind. It lacks some features that are present in the standard and data center editions. In addition, there is a maximum of twenty-five people or fifty devices, and client access licenses are not required.

Microsoft Server Standard

Microsoft Server Standard was designed for medium-to-large-sized companies with limited virtualization needs. Depending on the number of available cores, there are restrictions on the number of virtual machines that can be built and operated. Two per sixteen core licenses is the limit. In addition, the capacity to replicate data is restricted to 2 TB. In contrast to the fundamentals, client access licenses are necessary for users within the local network to access and utilize the server's services.

Microsoft Server Datacenter

Microsoft Server Datacenter and Microsoft Server Standard are comparable in many ways. This license allows for the creation of an infinite number of virtual machines. This license is structured in this manner to enable data centers to deliver this additional service. There is still a requirement for a client access license for hosts connecting to the server.

Per-Socket License

The Per-Socket License approach determines cost based on the number of processors in use. Each processor connects to a socket on the motherboard, from where the name is derived. It does not matter how many cores each processor has; what matters is the number of processors. The more processors, the faster the server, but the more expensive the license. This applies not only to physical servers but also to cloud virtualization. This is typically a hidden cost that, if not properly configured, can significantly increase the cost of the cloud infrastructure.

The intended function of the device should also be considered. For example, data backup does not require a great deal of processing power, but it does require greater bandwidth and network resources. If you have a backup server, having more sockets may not make sense.

Per-Core License

The Per-Core License is determined by the total number of cores housed within the CPU, regardless of the number of physical processors. Each of these cores can independently carry out its processing thread. For instance, a server with sixteen cores can simultaneously do sixteen distinct calculations. This results in faster processor processing speeds and various benefits, such as improved virtualization.

Per core licensing replaced per socket licensing since it provided a more accurate evaluation of processing capability and consistency across the different cloud platforms. It has enabled a greater number of enterprises to shift to the cloud as they see fit and allowed new businesses to invest in less infrastructure resulting in lower start-up costs.

Per-Instance License

A Per-Instance license is required for each physical server, virtual machine, or container that has the program installed. Per-instance licensing is the term used to describe this type of licensing. This may refer to the complete computer operating system, or it may refer to a specific program that the company utilizes. Because it only needs to be counted if it is installed on the machine, this kind of license is quite straightforward.

If a server or a virtual machine in an active/active configuration uses multiple copies of the software, each instance of that software needs to be counted. If a server or virtual machine is configured as an active/passive system, meaning that the secondary device is turned on only if the primary device becomes inoperable, just one license would be needed.

Volume Licensing

Volume Licensing is used to create ease of management for businesses, education, and government needs. Not to be confused with Site Licensing, this travels throughout the entire organization and is not restricted to a single site. Volume Licensing is likely to be used with operating systems and applications. It is convenient because the license will allow one software license key to be used for the specified number of devices. This makes administration much easier, not having to track the license key to each specific machine.

There are financial benefits that come with volume licensing as well. More accurate budgeting and forecasting of needs reduce costs. Also, some vendors will offer discounts and other benefits built into their program. Overall, this licensing style helps medium and large organizations stay current on their software licensing.

Concurrent License

The number of Concurrent Licenses required depends on how many users the software can support concurrently. When people work different shifts or in different geographical places and their times do not overlap, these licenses can be helpful.

For example, a company might buy ten licenses for the first shift to use while they are in the building. Once the first shift is over and the second shift workers come in, they could use the same licenses to connect if there were not more than ten people connected at the same time. Another case would be if a company did business all over the world. Because different countries work at various times, people in one country can use them during their work shifts. Since their shifts do not overlap, users on the other side of the world who log in for the day could use the same license once their shift is over. This can help cut costs and make it easier to keep track of licenses.

Node-Locked License

A Node-locked License is a type of license that is tied to specific hardware. Users will buy a license for software; sometimes, it may be that only one device can use it, or it may be for several devices. Once this is determined, the software will keep track of the hardware on which it is installed. For instance, if there were five available licenses for a program, it could be installed on five devices. The hardware and software will be connected via a unique identifier. If a user attempted to install the program on a sixth device, there would be no license to create the identifier, the hardware would not be authorized, and the software would not be allowed to install. If the user stops using a device, the license could be deactivated and transferred to a different device. This type of license makes it easy to manage because there is no need for a license server, and it helps developers crack down on sharing their software and unauthorized use.

Open Source License

An Open Source License is a type of license that makes the software's source code available to the end user. This idea was spearheaded by computer programmer Richard Stallman and the creation of the open-source initiative. The thought is that the source code to the software should be made available as it will lead to continued improvements and advancements in the computing field. The open-source initiative provides the structure for how the open-source community works. The source code is stored in public repositories that anyone can access. Users can then modify the coding of the software to add or remove features, change the way the software behaves, or create something completely custom to their purpose within the rules of the distribution license.

Operating systems like Windows and Mac do not share their source code, so they are called closed-source or proprietary. An example of an open source operating system is Linux. The open-source concept used by Linux allowed users to make custom versions of the operating system and share them with the public. A big draw to open source is that it is usually free or much cheaper than proprietary versions. This lowered cost does have drawbacks, though. Unlike paid software, not all open-source software developers provide technical support for their products. It will be the responsibility of the user to fix bugs and create security patches to fix the software, which will require a higher level of expertise.

Another advantage of open source is that switching to a different product or service is easier because you will not have the initial cost with closed sources. Some other benefits to open source would be more eyes collaborating on security, the lower total cost of ownership, large community sharing, increased innovation, and interoperability of other open-source software. Some cons would usually be no warranty, interoperability with closed source software, not usually used for business purposes, especially banking, and it is harder to use for most end users. Closed source software is more refined for the end user because they want them to purchase their product, whereas that is often not the goal of open source.

Subscription License

A Subscription-Based License has a price associated with its use over a specific period. Numerous services have offered subscriptions for a long time, but their popularity is growing as cloud computing becomes more widespread. The perpetual paradigm, in which software is purchased and retained indefinitely, is being replaced.

Why would someone prefer a subscription over ownership? Typically, the cost to begin utilizing the software will be lower initially. Typically, there is no setup price, and users pay monthly or annual fees to maintain access to the program and service. The subscription fee further covers technical assistance and software updates. Even if the user owns the program in the perpetual model, it will typically need to be changed within a few years due to hardware upgrades, end-of-life support, and other compatibility issues as technology advances. The subscription license intends to overcome this issue by incorporating future updates, support, cheaper upfront costs, and new feature development into the subscription cost.

Another advantage of adopting subscription licensing is greater flexibility. If you need additional licenses for a brief period, such as during peak season, you can purchase them for that period at a lower rate than if you bought them altogether.

License Vs. Maintenance and Support

A License for software gives the end user the right to use the software through the end-user license agreement. Licenses can help to ensure that unlicensed copies of the software are not being used and pirated. It specifies who controls what and how the software can be used.

The software vendor will usually provide software maintenance upon purchasing software through perpetual ownership or through software licensing. This would include security and patches, driver updates, and upgrades that can add additional features. Levels of service will vary, and often support will only be provided for the most current version of the unchanged software, and support may eventually come to the end of life.

Another form of support would be technical support. This is where users have issues and contact a helpdesk representative for assistance. Depending on the Service License Agreement for the software, there are differing levels of service and differing lengths of service. By having a subscription license, continued support is expected for products.

License Count Validation - True Up

License Count Validation - True Up is a yearly practice for businesses using an **Enterprise Agreement**. Depending on the level of service that is subscribed to will determine the resources being charged for. Businesses are then required to take an accurate inventory of their software licensing and make the required purchases. Below are a few things that are commonly checked to verify compliance:

- How many users are there compared to the previous estimate?
- Did the business increase computing power?
 - How many servers & cores on the hardware and virtual servers are there compared to the previous estimate?
 - Will clustering be deployed?
 - Are there any new end-user computers, whether hardware or virtualized and how many cores?
- Check for and document new subscription services like Microsoft Office and missing licenses.
- Verify operating systems licensing, application licensing, and client access licenses and blacklist any applications that shouldn't be used any longer.

For example, an organization is expected to have 1,000 qualified users and devices. After completing their true, they realize they have 1100 users and devices. To true up the licenses,

they would need to order one hundred Client Access Licenses for server access, 100 operating system licenses, and the rights to any other applications.

Exercise 2 - Version Compatibility

Technology is always changing and updating. Version compatibility aims to ensure that the software or hardware that you purchase today can still be used with previously released or upcoming releases of products. This help to keep costs in line and ensure that there are no interruptions in business costing money.

In this exercise, the concepts of backward compatibility and forward compatibility will be discussed.

Learning Outcomes

After completing this exercise, you should have further knowledge of:

- Backward Compatibility
- Forward Compatibility

Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

Backward Compatibility

Backward compatibility is updated software or hardware that will still be compatible with older hardware or previous software versions of programs and operating systems. This allows users and businesses to update but still achieve functionality and keep costs in line by using older hardware and software and not having to replace everything. Sometimes there is also a performance benefit like newer, faster hardware executing the older software faster and more efficiently.

Backward compatible is a common term used in many instances in the industry. When a new gaming system is released, backward compatible models will allow the users to still use their games from the previous system. Another example of this happening would be a business having a program written specifically for Windows 7 working with Windows 10 in compatibility mode.

Yet another example of this would be a program written for a thirty-two-bit operating system working on a sixty-four-bit operating system. Some drawbacks are common with backward compatibility. There can be increased costs due to complexity and security. Typically, older software will have bugs and loopholes that threat actors can exploit. It will be the organization's responsibility to supply fixes for these issues requiring a higher level of expertise to keep them operational.

Forward Compatibility

Software or hardware with forward compatibility was developed to ensure that it will continue functioning properly even after newer software versions or upgrades have been released. In almost all cases, any hardware or software that is forward compatible will also be backward compatible.

A frequent example of forward compatibility is additional features for a product still in development but provided as an update in the software or firmware. This is a common occurrence. The Apple iPhone is an excellent example of how forward compatibility can be implemented. Once a year, a new iPhone model with updated capabilities is released. The older versions of the phones have also had their software upgraded, and if they are capable, they will be able to use the new capabilities just like the newer ones.

Review

Well done, you have completed the **Server Licensing Concepts** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Licensing Models
- Exercise 2 - Version Compatibility

You should now have further knowledge of:

- Physical Vs. Virtual
- Site License
- Per-Server License
- Per-Socket License

- Per-Core License
- Per-Instance License
- Volume Licensing
- Concurrent License
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Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.